

Stage 3: Database Design -- Updated Data Summary

Formula 1 World Championship from Vopani “Rohanrao”

Sourced from Kaggle <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>

The dataset chosen by our group contains data on the Formula 1 (F1) World Championship from 1950 to 2024. F1 is a world-renowned auto-racing forum, and is widely considered a premier league for circuit racing. The world championships considered in this dataset are entire seasons appearing in any given year. One season comprises a series of races that takes place on distinct circuits and public roads across the world. The collection of races in sequence is referred to as “Grand Prix.”

Data contained in the file(s)

The contents of this dataset are measured from the World Championships mentioned before. Specifically, the data consists of tables pertaining to the following:

- races
- drivers
- constructors
- qualifying sessions
- circuits
- lap times
- the seasons and championships overall

Constructors are the designers and builders of the cars used in races. Some additional supporting tables are driver standings, constructor standings and results, as well as sprint results. Sprints or sprint races are shorter than circuits and do not require any driver to stop for pit stops. The dataset is split into distinct files consisting of tables for the above data, and so we may prescribe entities using those tables. Then, much of the relevant data from the tables can be considered attributes, which we describe below. A last file designated “status” simply carries a mapping of various statuses describing different entities.

Defining entities using the dataset

We have translated many of the tables directly to entities (e.g drivers.csv to the entity Driver). In the cases where this didn't make sense, we have instead chosen to move data either into relationships (e.g. putting data from qualifying.csv into the relationship "Races in") or have elected to simply ignore it as it is not relevant to the queries we intend to write (e.g. choosing not to include data from pit_stops.csv).

We have elected to keep the IDs as assigned in the data for all our entities. For the entities Driver, Constructor, Circuit, Lap, and Season we have pulled all columns from the respectively named table in our dataset as attributes, except in the following cases: the column stored data that would either be unnecessary for the queries we have planned, the data would already be available through a relationship, or each record was mainly null. Certain columns had over 90% of their records as null, so there is no sense in keeping them. This is discussed more below in relation to cleaning the data. For the entity Race, we did similarly, though we added a Boolean value that determines if a certain Race is a Grand Prix or a sprint. This information is from sprint_results.csv.

All of the results and data related to them (both for individual races and the seasons overall) are stored in relationships between Race and Driver, and Season and Driver/Constructor. These are either pulled from multiple tables (e.g. constructor_results.csv, results.csv) in the case of Races, or are derived from the points gained in Races (in the case of Seasons).

On the size of the dataset, and cleaning efforts

Across all files and entity sets, there is a total number of 701,433 records. The entity, lap times, accounts for approximately 500,000 records—reflecting the extensive number of tracked laps across many races and seasons. Excluding the quantity of records contributed by lap times, the remaining records come from constructor results and standings, driver standings, pit stops, qualifying races, and the results, with each having greater than 10,000 records respectively. The remaining entity sets—circuits, constructors, drivers, races, seasons, sprint results, and status—each have between 70 to 1,000 records. Dataset cleaning will be necessary due to the presence of NULL values in primary attributes, such as race lengths. We can omit less-essential attributes since querying for such offers limited insight, but total race lengths can be derived from the lap times of the last-place driver for any race. This would provide an estimate when the race should end, assuming the race did not end due to time constraints. Fortunately, there is a lot of data to work with in this dataset—conveniently allowing us to remove any attributes that are riddled with NULL values and no sensible default value to replace them with. To this end, we may write a python script to simply remove any columns with NULL values. Otherwise, for attributes that can support a working default value, we can simply replace any NULL values with our chosen default. Creating a script to do this is straightforward. In order to start with the process, we will create a separate script to discover attributes that contain NULL values. Upon the collection of such attributes, we will decide on an individual basis whether or not we should omit the attribute or substitute in a valid default value—if available.